Vishwjeet Chavan

Creative Technologist • Spatial Frontend Engineer • Interactive Designer

Portfolio: portfolio-spatial-eight.vercel.app
GitHub: github.com/vishwjeetlol
Location: India • Global Remote

✦ PROFESSIONAL PROFILE

Independent creative developer and spatial engineer pioneering real-time WebGL, custom GLSL shaders, procedural Web Audio synthesis, and physics-driven spatial interfaces. Proven track record architecting 12 uncompromised interactive systems running locked at 60 FPS with sub-second production builds (390ms). Blends vibrant cultural maximalism with rigorous computer graphics engineering.

✦ CORE TECHNICAL COMPETENCIES

Real-Time Graphics & 3D
Three.js (r160), WebGL, GLSL Shaders, Raymarched SDFs, GPGPU Curl Particles, Blender (Python headless pipeline).

Spatial Frontend & Systems
TypeScript, Modern JavaScript (ESM), Vite, HTML5 Canvas, CSS 3D Transforms, Web Audio API DSP, Playwright.

Performance & Math
60 FPS Optimization, Verlet Physics Solvers, Rigid-Body Impulses, Distance Culling, Zero-Allocation Vector Math.

✦ FLAGSHIP INTERACTIVE SYSTEMS & CREATIVE ENGINEERING

Liquid Chrome Simulation Lab // GLSL Fluid Dynamics — 2026

`WebGL • Raw GLSL Vertex Displacement • Split-Channel Fresnel Optics • Web Audio API`

- Engineered real-time fluid surface tension wave propagation using multi-octave simplex noise in custom GPU vertex shaders.
- Implemented split-channel RGB refraction caustics and coupled wave impulses to procedural audio transients.

Physics Playground Engine // 3D Rigid-Body Dynamics — 2026

`Three.js • Scratch Physics Solver • Spatial Hash Partitioning • Zero-Allocation Math`

- Developed a custom 3D rigid-body impulse engine with friction restitution manifolds, gravity vector fields, and sphere-box collision detection.
- Achieved sustained 60 FPS during 50+ body stress tests through spatial grid broad-phase acceleration.

Bioluminescent Swarm // 16,384 GPGPU Particle Field — 2026

`WebGL • GPGPU Shaders • 3D Curl Noise • Lorenz Attractor Topology`

- Built an interactive 16K particle system executing 3D curl noise turbulence and dynamic mathematical attractor morphing directly on GPU shaders.
- Coupled particle velocities and color dispersion to real-time Web Audio FFT frequency bands.

Cyber Tunnel 4D // Volumetric Raymarched SDFs — 2026

`Fragment Shader Sphere Tracing • Signed Distance Fields • Non-Euclidean Gyroids`

- Authored a single-pass volumetric raymarching engine rendering infinite non-Euclidean geometric corridors with camera mouse inertia.
- Engineered procedural surface displacement modulated dynamically by low-frequency audio energy.

Vinyl Turntable Studio // Real-Time Audio DSP & Scratch Physics — 2026

`Web Audio API • Procedural Sound Synthesis • Rotational Inertia Physics • Canvas 2D`

- Built a tactile vinyl turntable simulating physical needle drag, rotational friction, and real-time bidirectional scratch playback.
- Synthesized procedural rhythmic bass grooves and tone sweeps entirely in-browser without external audio files.

Cloth & Origami Simulation // Verlet Constraint Solver **2026**

`Three.js • Verlet Integration • Distance Constraints • Wind Dynamics`

- Implemented a real-time cloth physics simulation with structural, shear, and bending spring constraints, wind fields, and vertex dragging.
- Created spatial origami folding kinematics with dynamic lighting normals and backface rendering.

Experimental Systems Suite (Labs 07 – 12) // Optics & Intelligence **2026**

`Material Optics (Headless Blender) • Glass UI Suite • Motion Studies • Neural Playground • Seismic 3D • Digit Atlas`

- **Material Optics & Glass UI:** Headless Blender procedural caustics pipeline and 6-component spatial CSS 3D spring-physics UI kit.
- **Neural Systems & Data:** In-browser MLP neural net with live backprop decision boundary and 30-day USGS 3D planetary seismic globe.

✦ EDUCATION & BACKGROUND

Bachelor of Technology • Computer Science & Engineering **2026 – 2030**

Indian Institute of Information Technology, Dharwad (IIIT Dharwad)

Specialization in Computer Graphics, Numerical Algorithms, Spatial User Interfaces, and Real-Time Systems.
